

Rate, Leakage, and Distortion for Gaussian Sources with an Informed Encoder

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Abstract

An encoder observes a jointly Gaussian vector and describes part of it to a decoder under a weighted quadratic distortion. A second party holds a noisy linear observation of the source and never decodes. One description then carries two costs: its rate, and its conditional leakage to the second party, which is also the least rate at which that party can re-encode it. The single-letter region is known from secure source coding, and we determine its structure. The direction an optimal description reads turns with the distortion budget unless the decoder's and the second party's geometries share it, while every description built for a single party reads fixed directions, and a fixed read pays an explicit excess leakage. When the optimal description is one-dimensional, which an explicit test certifies, the minimum leakage is a closed-form function of the largest generalized eigenvalue of a matrix pencil in which the budget enters. Both costs are convex in the error covariance, so each frontier point solves a convex program. The frontier is explicit at every budget in commuting geometry and at small budgets in any geometry when the whole source is described, and an optimal description never needs more dimensions than the described variable. The tradeoff exists only because the distortion is weighted or partial. With decoder side information, both degraded cases are explicit.

Index Terms

Equivocation, Gaussian sources, information leakage, rate–distortion theory, secure source coding, side information.

I. INTRODUCTION

One description of data is often held by more than one party. The decoder that reconstructs from it pays its rate. A second party that holds the description together with its own observation of the source learns from it the conditional mutual information between the source and the description given that observation. In a secrecy reading the second party is an eavesdropper and this quantity is the information leakage beyond its side information; minimizing it maximizes the eavesdropper's equivocation about the source. In a storage reading the second party keeps the description compressed against what it already knows, and the same quantity is the least rate at which it can re-encode the description. Either way one description carries two costs, set by two different parties, and the encoder, which may observe more of the source than either party, decides how they trade off.

The single-letter region of jointly achievable rate, distortion, and equivocation is due to Villard and Piantanida [1, Theorem 3], and for vector Gaussian sources Ekrem and Ulukus [2] obtained the minimum leakage when the rate is unconstrained, as an optimization over covariance matrices. The joint rate–leakage region was left open: they report that they could not solve the weighted problem that traces it [2, p. 8], and Xu and Chen [5] have since proved by a new extremal inequality that Gaussian auxiliaries solve it, leaving the region as a matrix program. Without decoder side information the region itself is elementary, because a single Gaussian exhaustion lemma bounds both costs at once (Proposition 2). What this paper contributes is the structure of that region for an informed encoder. The encoder observes a jointly Gaussian vector T , the decoder must reproduce a linear part $Y = FT$ under a weighted quadratic distortion, and the second party observes $S = HT + U$ for an arbitrary matrix H . When H reaches the parts of T outside Y , the encoder holds information about the second party's observation that the decoder does not need, and this is what makes the two costs disagree. The disagreement also needs a weighted or partial distortion. Under a matrix distortion on the whole source, the formulation of the earlier work, one description minimizes both costs (Proposition 4).

The results are structural as well as explicit. The coding theorem (Theorem 3) identifies the operational region with the single-letter one, and Gaussian descriptions, indexed by their error covariance, exhaust it (Proposition 2). In that coordinate both costs are convex (Theorem 5), so every point of the rate–leakage frontier is the unique minimizer of a convex program and the minimum leakage is a determinant maximization under linear matrix inequalities. When the optimal description is one-dimensional, which an explicit certificate decides, the minimum leakage is

$$L(D) = \frac{1}{2} \log(1 + \Delta/\mu_{\max}),$$

with Δ the reduction in distortion the description must buy and $\mu_{\max}$ the largest generalized eigenvalue of a pencil formed from the source covariance, the decoder’s weight, the conditional covariance left after the second party’s observation, and the budget (Theorem 11). For a scalar described variable and a scalar context this is an explicit closed form (Corollary 12). When the decoder’s and the holder’s geometries commute, the whole frontier is explicit at every budget (Theorem 8), and more generally the problem splits along every subspace decomposition that both geometries respect (Theorem 7). When the decoder reproduces the whole source, the frontier is explicit at every small budget in any geometry, in the eigenbasis of the second party’s information relative to the decoder’s weight (Theorem 10). In general an optimal description has no more components than the described variable, even though it uses the context (Proposition 6).

The eigenvector of the pencil is the direction the description reads, and it carries the paper’s main structural finding. The rate-optimal description reads the principal direction of the decoder’s weighted covariance at every budget, as do principal components, reverse water-filling [3], and the Gaussian information bottleneck [4]. The leakage-optimal description reads a direction that turns with the budget, unless that direction is a common generalized eigenvector of the decoder’s geometry and the second party’s (Theorem 14). No description optimized for a single party does this (Proposition 13), and a description that reads a fixed direction pays an explicit excess leakage, at one of any two budgets whose optimal reads differ (Theorem 19). The same condition governs whether the two optimal descriptions coincide at all (Theorem 16): in the one-dimensional regime they differ whenever the second party’s information does not map the decoder’s principal direction onto itself. For independent components the leakage-optimal split of the distortion equalizes slopes that we compute in closed form, activates components in an order that is not the order of their variances, and can describe a disjoint set of components from the rate-optimal split (Section VII). With side information at the decoder (Section VIII) the two degraded cases are explicit: when the decoder’s information dominates the holder’s, the leakage equals the rate at the optimum and there is no tradeoff; when the holder’s dominates, an estimation inequality (Lemma 23) reduces the problem exactly to the one without decoder side information, for the source conditioned on the decoder’s observation and the holder’s excess information. The same effects appear in fixed-rate scalar quantizers (Section X). At equal rate and equal distortion, a Lloyd–Max quantizer applied to a leakage-aware projection leaks 13% to 72% less than the same quantizer on the principal axis, and the projection it reads turns with the budget.

Related work

The conditional leakage studied here is the secrecy literature’s quantity. For a deterministic encoder, $h(T^n | M, S^n) = h(T^n | S^n) - H(M | S^n)$, so minimizing $H(M | S^n)/n$, which equals $I(T^n; M | S^n)/n$, is the same as maximizing the equivocation $h(T^n | M, S^n)/n$. Secure lossy source coding with side information at the eavesdropper was characterized by Villard and Piantanida [1]; with no decoder side information their Theorem 3 gives the discrete region of Theorem 26 below. Ekrem and Ulukus [2] treat vector Gaussian sources with side information at both receivers under a matrix distortion on the whole source, give an outer bound that is tight without a rate constraint, and leave the joint region open. Our distortion constrains a weighted part of the source and leaves the rest unpenalized, the decoder has no side information, and the eavesdropper’s observation map is arbitrary. Xu and Chen [5] prove, by a new extremal inequality, that Gaussian auxiliaries are optimal in the Ekrem–Ulukus problem for every weight, which characterizes the rate–distortion–equivocation function as a matrix program over two precision matrices [5, eq. (189), Thm. 4]. Their model has an eavesdropper observation $X + N_Z$ with $\text{Cov } N_Z \succ 0$, which excludes a rank-deficient observation such as a scalar context, and a matrix distortion on the whole source. In the limit of no decoder side information their program reduces to the one of Proposition 2 in the precision coordinate $\Sigma_{e0}^{-1} - \Sigma_T^{-1}$, with the matrix constraint in place of the weighted one, and then it has no tradeoff (Proposition 4). Their result is a converse that fixes the form of the optimizers; the structure of the

solution, which is the subject here, is not addressed there. Günlü, Schaefer, Boche, and Poor [6] evaluate the lossy rate region for scalar Gaussian sources with a remote source and separate secrecy and privacy leakage. Neither treats an encoder that observes components it need not describe, a weighted partial distortion with an arbitrary observation map, or the structure results below. Related problems price the leakage of a description about side information [7], [8], [9] or the equivocation of the source in a cipher system [10], [11]. The leakage endpoint is Steinberg's common-reconstruction functional [12] on the source as the encoder observes it, an augmentation noted for finite alphabets by Lapidot, Malär, and Wigger [13]; a vector Gaussian common-reconstruction function with every component penalized was given by Lu and Xu [14]. Weighted quadratic distortion models a decoder that cares about some directions more than others, as in weighted rate–distortion [15], indirect source coding [16], and task-based quantization [17]. Parameterizing a Gaussian description by its error covariance is a standard device in vector Gaussian source coding [18], and the resulting endpoint program is a determinant maximization [19].

The paper is organized as follows. Section II sets up the model and the coding theorem, Section III proves the convex structure, Section IV decoupling and commuting geometry, Section V the one-dimensional regime, Section VI the results on directions and misalignment, Section VII independent components, and Section VIII decoder side information. Section IX gives instances, Section X evaluates practical scalar quantizers, and Section XI open problems. The appendices prove the coding theorem and a Gaussian exhaustion lemma and describe the numerical checks.

II. MODEL AND THE CODING THEOREM

Logarithms are base two unless written $\ln$. The encoder observes i.i.d. copies of $T \in \mathbb{R}^p$, jointly Gaussian and centered with $\Sigma_T \succ 0$. The described variable is $Y = FT \in \mathbb{R}^m$ for a matrix F of full row rank, and the decoder's distortion is $\mathbb{E}[(Y - \hat{Y})^\top W(Y - \hat{Y})]$ for a weight $W \succ 0$. Write

$$Q := F^\top W F \succeq 0, \quad D_{\max} := \text{tr}(Q \Sigma_T).$$

The second party, which we call the holder, observes $S = HT + U \in \mathbb{R}^r$ with $U \sim \mathcal{N}(0, \Sigma_U)$, $\Sigma_U \succ 0$, independent of T and of all coding randomness. Write

$$J := H^\top \Sigma_U^{-1} H \succeq 0, \quad \Sigma_{T|S} = (\Sigma_T^{-1} + J)^{-1},$$

the information that S carries about T and the conditional covariance it leaves.

A code consists of an encoder $f_n : \mathbb{R}^{p \times n} \rightarrow \mathcal{M}_n$, a finite set, and a decoder $g_n : \mathcal{M}_n \rightarrow \mathbb{R}^{m \times n}$. With $M = f_n(T^n)$ and $\hat{Y}^n = g_n(M)$, its rate, distortion, and leakage are

$$R_n = \frac{1}{n} \log |\mathcal{M}_n|, \quad D_n = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[(Y_i - \hat{Y}_i)^\top W(Y_i - \hat{Y}_i)], \quad L_n = \frac{1}{n} H(M | S^n) = \frac{1}{n} I(T^n; M | S^n).$$

A pair (R, L) is achievable at distortion D when a sequence of codes has $\limsup R_n \leq R$, $\limsup L_n \leq L$, and $\limsup D_n \leq D$; the closure of the achievable set is $\mathcal{R}(D)$. Throughout $0 < D < D_{\max}$; at $D \geq D_{\max}$ the zero reproduction gives $R = L = 0$. Write $R_{\min}(D)$ and $L(D)$ for the least rate and the least leakage over $\mathcal{R}(D)$, and call a description rate-optimal or leakage-optimal when it attains one of them.

A test channel $p(\hat{y} | t)$ is admissible when its distortion is at most D ; the Markov chain $\hat{Y} - T - S$ holds because the description is formed without S . For a channel, let Σ_{e0} and Σ_e be the error covariances of the best linear estimates of T from $\hat{Y}$ and from $(\hat{Y}, S)$. Appendix B proves the following Gaussian exhaustion lemma.

Lemma 1. *Let $\hat{Y} \in \mathbb{R}^k$, for any k , be centered with finite second moments, let A be its regression matrix on T and $N' = \hat{Y} - AT$ its residual, and suppose $\text{Cov}(N', S) = 0$ and $\text{Cov } N' \succ 0$. Then $I(T; \hat{Y}) \geq \frac{1}{2} \log(\det \Sigma_T / \det \Sigma_{e0})$ and $I(T; \hat{Y} | S) \geq \frac{1}{2} \log(\det \Sigma_{T|S} / \det \Sigma_e)$, and the jointly Gaussian channel $AT + N$, $N \sim \mathcal{N}(0, \text{Cov } N')$ independent of (T, U) , has the same second moments and meets both bounds with equality.*

Proposition 2. *The union of the quadrants above $(I(T; \hat{Y}), I(T; \hat{Y} | S))$ over admissible channels equals the union, over $\Sigma_{e0} \in \mathcal{P}(D)$, of the quadrants above*

$$(r(\Sigma_{e0}), \ell(\Sigma_{e0})) = \left(\frac{1}{2} \log \frac{\det \Sigma_T}{\det \Sigma_{e0}}, \frac{1}{2} \log \frac{\det \Sigma_{T|S}}{\det \Sigma_e} \right), \quad \Sigma_e^{-1} = \Sigma_{e0}^{-1} + J, \quad (1)$$

where $\mathcal{P}(D) = \{\Sigma_{e0} : 0 \prec \Sigma_{e0} \preceq \Sigma_T, \text{tr}(Q\Sigma_{e0}) \leq D\}$.

Proof. Given an admissible channel, centering $\hat{\mathbf{Y}}$ changes neither coordinate and does not raise the distortion. If $\text{Cov } N'$ is singular, either a noiseless linear functional of T is revealed and both coordinates are infinite, or a nonzero combination of the components of $\hat{\mathbf{Y}}$ vanishes and may be dropped after an invertible linear map; so assume $\text{Cov } N' \succ 0$. The residual is uncorrelated with T by construction and with U by independence, hence with S , and Lemma 1 replaces the channel by $\hat{\mathbf{Y}} = AT + N$ with the same distortion and no larger coordinates. For it, $\Sigma_{e0}^{-1} = \Sigma_T^{-1} + A^\top \Sigma_N^{-1} A$, so $0 \prec \Sigma_{e0} \preceq \Sigma_T$; the pair $(\hat{\mathbf{Y}}, S)$ is a linear observation of T with independent noises, so $\Sigma_e^{-1} = \Sigma_{e0}^{-1} + J$; and the distortion is at least that of $\mathbb{E}[Y | \hat{\mathbf{Y}}]$, which is $\text{tr}(Q\Sigma_{e0})$. Conversely, given $\Sigma_{e0} \in \mathcal{P}(D)$, factor $\Sigma_{e0}^{-1} - \Sigma_T^{-1} = G^\top G$, let $Z = GT + N$ with $N \sim \mathcal{N}(0, I)$ independent of (T, U) , and reproduce $\hat{\mathbf{Y}} = \mathbb{E}[Y | Z]$. Then $\text{Cov}(T | Z) = \Sigma_{e0}$, the distortion is $\text{tr}(Q\Sigma_{e0}) \leq D$, $\hat{\mathbf{Y}} - T - S$ holds, and since $\hat{\mathbf{Y}}$ is a function of Z , data processing bounds its coordinates by those of Z , which are (1). $\square$

Theorem 3. For every $0 < D < D_{\max}$, $\mathcal{R}(D)$ equals the union of Proposition 2.

The proof is in Appendix A. Its discrete core is the region of Villard and Piantanida without decoder side information (Theorem 26); the Gaussian extension quantizes the side information, the source, and the reproduction and passes to the limit.

The tradeoff studied below needs the distortion to leave the encoder a choice of error covariance. The vector Gaussian secrecy literature [2], [5] instead asks the decoder to reproduce the whole source under a matrix constraint, and without decoder side information that choice disappears.

Proposition 4. Suppose the decoder reproduces all of T , an admissible channel is one whose reproduction $\hat{T}$ has $\mathbb{E}[(T - \hat{T})(T - \hat{T})^\top] \preceq \Delta$, and $0 \prec \Delta \preceq \Sigma_T$. Then the union of the quadrants above $(I(T; \hat{T}), I(T; \hat{T} | S))$ over admissible channels is the single quadrant above $(r(\Delta), \ell(\Delta))$ of (1), so one description is both rate-optimal and leakage-optimal.

Proof. The proof of Proposition 2 applies with the matrix constraint in place of the trace, because the error matrix of $\hat{T}$ depends only on second moments and dominates $\text{Cov}(T | \hat{T}) = \Sigma_{e0}$ for the Gaussian channel. The feasible set becomes $\{0 \prec \Sigma_{e0} \preceq \Delta\}$. Both r and ℓ are nonincreasing in the Loewner order, since $\Sigma_e = (\Sigma_{e0}^{-1} + J)^{-1}$ is monotone in Σ_{e0} , and Δ is the largest feasible element. $\square$

Under the weighted constraint $\text{tr}(Q\Sigma_{e0}) \leq D$ the feasible set has no largest element, and which maximal element the encoder picks is the whole problem. The same is true of a matrix constraint on the described part Y alone, $F\Sigma_{e0}F^\top \preceq \Delta_Y$, which leaves the error covariance free outside Y .

III. CONVEX STRUCTURE

Theorem 5. On $\{0 \prec \Sigma_{e0} \preceq \Sigma_T\}$, the rate r and the leakage ℓ of (1) are strictly convex functions of Σ_{e0} , and the distortion $\text{tr}(Q\Sigma_{e0})$ is linear. Consequently:

- (a) for every $\alpha \in [0, 1]$ the program $\min\{\alpha r(\Sigma_{e0}) + (1 - \alpha)\ell(\Sigma_{e0}) : \Sigma_{e0} \in \mathcal{P}(D)\}$ has a unique minimizer, the distortion constraint is active there, and the minima over α trace the Pareto frontier of $\mathcal{R}(D)$;
- (b) in the coordinate Σ_e the minimum leakage is a determinant maximization,

$$L(D) = \frac{1}{2} \log \det \Sigma_{T|S} - \frac{1}{2} \max\{\log \det \Sigma_e : 0 \prec \Sigma_e \preceq \Sigma_{T|S}, \text{tr}(Qh(\Sigma_e)) \leq D\}, \quad h(\Sigma_e) := (\Sigma_e^{-1} - J)^{-1}, \quad (2)$$

in which h is matrix-convex and the constraint is equivalent to a linear matrix inequality: with $J = B^\top B$, $\text{tr}(Qh(\Sigma_e)) \leq D$ if and only if some symmetric Z has $\text{tr}(QZ) \leq D$ and

$$\begin{bmatrix} Z - \Sigma_e & \Sigma_e B^\top \\ B \Sigma_e & I - B \Sigma_e B^\top \end{bmatrix} \succeq 0; \quad (3)$$

- (c) a feasible Σ_e is the maximizer in (2) if and only if $\text{tr}(Qh(\Sigma_e)) = D$ and there are $\varpi \geq 0$ and $\Theta \succeq 0$ with

$$\frac{1}{2} \Sigma_e - \varpi \Sigma_{e0} Q \Sigma_{e0} = \Sigma_e \Theta \Sigma_e, \quad \Theta(\Sigma_{T|S} - \Sigma_e) = 0, \quad \Sigma_{e0} = h(\Sigma_e), \quad (4)$$

and then $\varpi > 0$.

Proof. Convexity. $r = \frac{1}{2} \log \det \Sigma_T - \frac{1}{2} \log \det \Sigma_{e0}$ is strictly convex. For the leakage, write $J = B^\top B$ and note that $\Sigma_e = (\Sigma_{e0}^{-1} + B^\top B)^{-1}$ is the error covariance of the best linear estimate of a vector with covariance Σ_{e0} from its noisy image under B with unit noise. For every matrix L , the estimator with gain L has error covariance $E_L(\Sigma_{e0}) = (I - LB)\Sigma_{e0}(I - LB)^\top + LL^\top \succeq \Sigma_e$, with equality at the optimal gain, and E_L is affine in Σ_{e0} . A pointwise minimum of affine maps in the Löwner order, attained at each point, is matrix-concave: at $\Sigma_t = (1-t)\Sigma_0 + t\Sigma_1$ with optimal gain L_t , $\Sigma_e(\Sigma_t) = E_{L_t}(\Sigma_t) = (1-t)E_{L_t}(\Sigma_0) + tE_{L_t}(\Sigma_1) \succeq (1-t)\Sigma_e(\Sigma_0) + t\Sigma_e(\Sigma_1)$. The map $\Sigma \mapsto -\log \det \Sigma$ is strictly convex and decreasing in the Löwner order, so $-\log \det \Sigma_e(\Sigma_t) \leq -\log \det ((1-t)\Sigma_e(\Sigma_0) + t\Sigma_e(\Sigma_1)) \leq (1-t)(-\log \det \Sigma_e(\Sigma_0)) + t(-\log \det \Sigma_e(\Sigma_1))$, with strict inequality in the second step when $\Sigma_0 \neq \Sigma_1$, because $\Sigma_{e0} \mapsto \Sigma_e$ is injective. So ℓ is strictly convex.

(a) $\mathcal{P}(D)$ is convex, the objective is strictly convex and diverges as $\det \Sigma_{e0} \rightarrow 0$, and the feasible set is bounded, so the minimizer exists and is unique. Both r and ℓ decrease when Σ_{e0} increases in the Löwner order, so at a minimizer with slack distortion a move toward Σ_T would stay feasible and lower the objective, since Σ_T itself has distortion $D_{\max} > D$. The region is convex and upward closed, so its Pareto points are exactly the minima of the weighted programs.

(b) The map $\Sigma_{e0} \mapsto \Sigma_e$ is a bijection of $\{0 \prec \Sigma_{e0} \preceq \Sigma_T\}$ onto $\{0 \prec \Sigma_e \preceq \Sigma_{T|S}\}$, with inverse $h: \Sigma_{e0} \preceq \Sigma_T$ if and only if $\Sigma_e^{-1} = \Sigma_{e0}^{-1} + J \succeq \Sigma_T^{-1} + J = \Sigma_{T|S}^{-1}$. The matrix $B\Sigma_{T|S}B^\top$ has the same nonzero eigenvalues as $\Sigma_{T|S}^{1/2}J\Sigma_{T|S}^{1/2} \prec \Sigma_{T|S}^{1/2}(\Sigma_T^{-1} + J)\Sigma_{T|S}^{1/2} = I$, so on the feasible set $B\Sigma_eB^\top \prec I$, and the Woodbury identity gives $h(\Sigma_e) = \Sigma_e + \Sigma_eB^\top(I - B\Sigma_eB^\top)^{-1}B\Sigma_e$. By the Schur complement criterion [20, Sec. A.5.5], $h(\Sigma_e) \preceq Z$ if and only if (3) holds, and the block matrix there is affine in (Σ_e, Z) ; averaging two instances with $Z_i = h(\Sigma_e^{(i)})$ gives matrix convexity of h . Since $\text{tr}(QZ)$ is monotone in Z , the constraint $\text{tr}(Qh(\Sigma_e)) \leq D$ is equivalent to the existence of Z with $\text{tr}(QZ) \leq D$ and (3).

(c) The program (2) has a strictly feasible point ($\Sigma_{e0} = \epsilon\Sigma_T$ with $\epsilon D_{\max} < D$), so the Karush–Kuhn–Tucker conditions are necessary and sufficient. The differential of h is $dh = h\Sigma_e^{-1}d\Sigma_e\Sigma_e^{-1}h$, so the gradient of $\text{tr}(Qh)$ is $\Sigma_e^{-1}\Sigma_{e0}Q\Sigma_{e0}\Sigma_e^{-1}$, and stationarity of $-\frac{1}{2} \ln \det \Sigma_e + \varpi \text{tr}(Qh) + \text{tr}(\Theta(\Sigma_e - \Sigma_{T|S}))$, multiplied on both sides by Σ_e , is the first condition of (4), the factor $\ln 2$ being absorbed in ϖ . When the distortion constraint is active its complementary slackness holds for every $\varpi \geq 0$, and the maximizer has an active constraint by (a). If $\varpi = 0$, then $\Theta = \frac{1}{2}\Sigma_e^{-1} \succ 0$ and the second condition forces $\Sigma_e = \Sigma_{T|S}$, which is infeasible; so $\varpi > 0$. $\square$

The optimal description uses the context, but it never needs more dimensions than the described variable.

Proposition 6. *For every $\alpha \in [0, 1]$ the minimizer Σ_{e0} of Theorem 5(a) has $\text{rank}(\Sigma_T - \Sigma_{e0}) \leq \text{rank } Q$, so it is attained by a jointly Gaussian description $GT + N$ with at most $\text{rank } Q$ components.*

Proof. The weighted program in Σ_{e0} is convex with the strictly feasible point $\epsilon\Sigma_T$, so there are $\nu \geq 0$ and $\Theta \succeq 0$ with $\Theta(\Sigma_T - \Sigma_{e0}) = 0$ and, from the gradients $\nabla r = -\frac{1}{2}\Sigma_{e0}^{-1}$ and $\nabla \ell = -\frac{1}{2}\Sigma_{e0}^{-1}\Sigma_e\Sigma_{e0}^{-1}$ (in natural units),

$$\frac{1}{2}\Sigma_{e0}^{-1}(\alpha\Sigma_{e0} + (1-\alpha)\Sigma_e)\Sigma_{e0}^{-1} = \nu Q + \Theta.$$

The left side is positive definite, so $\text{rank } \Theta \geq p - \text{rank } Q$, and the range of $\Sigma_T - \Sigma_{e0}$ lies in the null space of Θ . The precision $\Sigma_{e0}^{-1} - \Sigma_T^{-1} = \Sigma_{e0}^{-1}(\Sigma_T - \Sigma_{e0})\Sigma_T^{-1}$ has the same rank, and factoring it as $G^\top G$ gives the description of Proposition 2. $\square$

IV. DECOUPLING AND COMMUTING GEOMETRY

Whiten by the source covariance: write $\tilde{Q} = \Sigma_T^{1/2}Q\Sigma_T^{1/2}$, $\tilde{J} = \Sigma_T^{1/2}J\Sigma_T^{1/2}$, and $X = \Sigma_T^{-1/2}\Sigma_{e0}\Sigma_T^{-1/2}$. Since $\Sigma_{T|S} = \Sigma_T^{1/2}(I + \tilde{J})^{-1}\Sigma_T^{1/2}$ and $\Sigma_{e0}^{-1} + J = \Sigma_T^{-1/2}(X^{-1} + \tilde{J})\Sigma_T^{-1/2}$, the costs and the constraint of Proposition 2 become

$$r = -\frac{1}{2} \log \det X, \quad \ell = \frac{1}{2} \log \det(X^{-1} + \tilde{J}) - \frac{1}{2} \log \det(I + \tilde{J}), \quad \text{tr}(\tilde{Q}X) \leq D, \quad 0 \prec X \preceq I. \quad (5)$$

Theorem 7. *Suppose $\mathbb{R}^p = \bigoplus_k \mathcal{V}_k$ is an orthogonal decomposition into subspaces invariant under both $\tilde{Q}$ and $\tilde{J}$. Then for every $\alpha \in [0, 1]$ the minimizer of Theorem 5(a), written in the coordinate X , is block-diagonal with respect to the decomposition, and the weighted program splits into one program per block, coupled only through an allocation $D = \sum_k D_k$ that equalizes the slopes of the blocks' optimal values.*

Proof. Let P_k be the orthogonal projector onto $\mathcal{V}_k$, and for signs $\epsilon_k \in \{\pm 1\}$ let $S_\epsilon = \sum_k \epsilon_k P_k$, an orthogonal symmetric matrix that commutes with $\tilde{Q}$ and $\tilde{J}$. The map $X \mapsto S_\epsilon X S_\epsilon$ preserves $\log \det X$, $\log \det(X^{-1} + \tilde{J})$ because $(S_\epsilon X S_\epsilon)^{-1} + \tilde{J} = S_\epsilon(X^{-1} + \tilde{J})S_\epsilon$, the distortion $\text{tr}(\tilde{Q} S_\epsilon X S_\epsilon) = \text{tr}(\tilde{Q} X)$, and the constraint $X \preceq I$. The minimizer is unique, so it is fixed by every S_ϵ , and then $P_k X P_l = \epsilon_k \epsilon_l P_k X P_l$ for all signs, which forces $P_k X P_l = 0$ for $k \neq l$. For block-diagonal X both costs and the distortion are sums over blocks, and the remaining problem is a separable convex program with one linear constraint, whose optimality condition is a common subgradient level. $\square$

The independent components of Section VII are the case of two-dimensional blocks, each holding one described coordinate and its context. When the blocks are one-dimensional, which happens exactly when $\tilde{Q}$ and $\tilde{J}$ commute, the whole region is explicit.

Theorem 8. Suppose $Q \Sigma_T J = J \Sigma_T Q$, equivalently $\tilde{Q} \tilde{J} = \tilde{J} \tilde{Q}$, and let $u_1, \dots, u_p$ be a common orthonormal eigenbasis with $\tilde{Q} u_i = q_i u_i$ and $\tilde{J} u_i = j_i u_i$. For $\alpha \in [0, 1]$ the frontier point of weight α is $X^* = \sum_i x_i u_i u_i^\top$, where $x_i = 1$ if $q_i = 0$ and otherwise $x_i = \min\{1, \xi_i(\theta)\}$, with $\xi_i(\theta)$ the positive root of

$$\theta q_i j_i x^2 + (\theta q_i - \alpha j_i) x - 1 = 0 \quad (6)$$

($\xi_i = 1/(\theta q_i)$ when $j_i = 0$), and $\theta > 0$ the unique level with $\sum_i q_i x_i = D$. Its rate and leakage are

$$R = \sum_i \frac{1}{2} \log \frac{1}{x_i}, \quad L = \sum_i \frac{1}{2} \log \frac{1 + j_i x_i}{x_i(1 + j_i)}.$$

In particular the minimum leakage takes $x_i = \min\{1, (\sqrt{1 + 4j_i/(\theta q_i)} - 1)/(2j_i)\}$, and direction i is described exactly when $\theta > (1 + \alpha j_i)/(q_i(1 + j_i))$, so the leakage-optimal description begins with the directions of largest $q_i(1 + j_i)$, the rate-optimal one with those of largest q_i .

Proof. By Theorem 7 with the one-dimensional blocks spanned by the u_i , the minimizer is $\sum_i x_i u_i u_i^\top$ and the program is $\min \sum_i \phi_i(x_i)$ subject to $\sum_i q_i x_i \leq D$ and $0 < x_i \leq 1$, with $\phi_i(x) = \frac{\alpha}{2} \ln \frac{1}{x} + \frac{1-\alpha}{2} \ln \frac{1+j_i x}{x(1+j_i)}$ in natural units. Each ϕ_i is convex and decreasing, with $-\phi'_i(x) = (1 + \alpha j_i x)/(2x(1 + j_i x))$. A direction with $q_i = 0$ does not enter the constraint, so $x_i = 1$. For the others the optimality condition is $-\phi'_i(x_i) = \frac{1}{2} \theta q_i$ when $x_i < 1$ and $-\phi'_i(1) \leq \frac{1}{2} \theta q_i$ when $x_i = 1$, for a common $\theta > 0$; the first is (6), whose roots have product $-1/(\theta q_i j_i) < 0$, so exactly one is positive, and the second reads $\theta \leq (1 + \alpha j_i)/(q_i(1 + j_i))$. Each $x_i(\theta)$ is continuous and nonincreasing, strictly decreasing where it is below one, so $\sum_i q_i x_i(\theta)$ decreases continuously and strictly from $\sum_i q_i > D$ and θ is unique. The formulas for R and L are (5) at a diagonal X . $\square$

In each direction the leakage $\frac{1}{2} \log[(1 + j_i x_i)/(x_i(1 + j_i))]$ is Steinberg's scalar Gaussian common-reconstruction function [12] at unit variance, distortion x_i , and squared correlation $j_i/(1 + j_i)$ between the direction and the holder's observation; the theorem is the allocation of the distortion budget among such problems, and the joint frontier that goes with it.

Corollary 9. In commuting geometry every optimal description, at every budget and weight, reads the fixed directions u_i , consistent with Theorem 14. The rate-optimal and leakage-optimal descriptions nonetheless differ whenever two directions described by the rate-optimal one have different ratios j_i/q_i .

Proof. The first statement is Theorem 8. If the two optimal descriptions coincided, they would describe the same directions with the same x_i ; for those directions $x_i = 1/(\theta_R q_i)$ and $x_i(1 + j_i x_i) = 1/(\theta_L q_i)$, so $1 + j_i x_i = \theta_R/\theta_L$ is the same for all of them, and $j_i x_i = j_i/(\theta_R q_i)$ forces j_i/q_i to be the same. $\square$

When the decoder reproduces the whole source, the same per-direction formula holds without commuting geometry at every small budget, in the basis of the pencil (J, Q) instead of the common eigenbasis.

Theorem 10. Suppose $Q \succ 0$, and let $P = Q^{-1/2} J Q^{-1/2} = \sum_i p_i w_i w_i^\top$ with orthonormal w_i . For $\alpha \in [0, 1]$ and $\theta > 0$ let $z_i(\theta)$ be the positive root of

$$\theta p_i z^2 + (\theta - \alpha p_i) z - 1 = 0 \quad (7)$$

($z_i = 1/\theta$ when $p_i = 0$), let $\theta(D)$ be the unique level with $\sum_i z_i = D$, and let $\Sigma_{e0}(D) = Q^{-1/2}(\sum_i z_i w_i w_i^\top)Q^{-1/2}$. Then $\Sigma_{e0}(D)$ is increasing in D in the Löwner order, the set of D with $\Sigma_{e0}(D) \preceq \Sigma_T$ is an interval $(0, D_{\text{int}}(\alpha)]$ with $D_{\text{int}}(\alpha) > 0$, and on it $\Sigma_{e0}(D)$ is the frontier point of weight α , with

$$R = \frac{1}{2} \log \det(Q\Sigma_T) - \sum_i \frac{1}{2} \log z_i, \quad L = \frac{1}{2} \log \det(Q\Sigma_{T|S}) + \sum_i \frac{1}{2} \log \frac{1 + p_i z_i}{z_i}.$$

No commuting assumption is made.

Proof. In the coordinate $Z = Q^{1/2}\Sigma_{e0}Q^{1/2}$ the distortion is $\text{tr } Z$, and $\Sigma_e^{-1} = Q^{1/2}(Z^{-1} + P)Q^{1/2}$, which gives the formulas for R and L at $Z = \sum_i z_i w_i w_i^\top$. When the constraint $\Sigma_{e0} \preceq \Sigma_T$ is slack, its multiplier vanishes, and the stationarity condition in the proof of Proposition 6, multiplied on the right by Σ_{e0} , reads $\alpha I + (1 - \alpha)(I + J\Sigma_{e0})^{-1} = 2\nu Q\Sigma_{e0}$, which in the coordinate Z is

$$\alpha I + (1 - \alpha)(I + PZ)^{-1} = \theta Z, \quad \theta = 2\nu.$$

For $\alpha < 1$ the right side is symmetric, so $I + PZ$ is symmetric and $PZ = ZP$; for $\alpha = 1$ the condition is $Z = I/\theta$. Either way Z is diagonal in the basis w_i , and on the eigenvalues the condition is $\alpha + (1 - \alpha)/(1 + p_i z_i) = \theta z_i$, which is (7). Its left side decreases and its right side increases in $z_i > 0$, so the positive root is unique and decreases in θ ; hence $\theta(D)$ is unique and each z_i increases in D , and so does $\Sigma_{e0}(D)$, by congruence. As $D \rightarrow 0$, $\Sigma_{e0}(D) \rightarrow 0 \prec \Sigma_T$. When $\Sigma_{e0}(D) \preceq \Sigma_T$, the point satisfies the Karush–Kuhn–Tucker conditions with zero multiplier for the cap, which suffice by convexity, and Theorem 5(a) makes it the unique minimizer. $\square$

When $Q\Sigma_T J = J\Sigma_T Q$ and no cap of Theorem 8 is active, the two results agree, with $z_i = q_i x_i$ and $p_i = j_i/q_i$. When Q is singular, Proposition 6 shows that the cap is always active, and the error covariance equals Σ_T on a subspace of dimension at least $p - \text{rank } Q$.

V. THE ONE-DIMENSIONAL REGIME

Write $\Delta = D_{\text{max}} - D$ for the reduction in distortion the description must buy, and

$$A(\Delta) = \Sigma_T Q \Sigma_T - \Delta \Sigma_T.$$

A one-dimensional description is $Z = b^\top T + N$ with $b \neq 0$ and $N \sim \mathcal{N}(0, \sigma^2)$ independent, reproduced by $\mathbb{E}[Y | Z]$. Its read direction is b , and the direction in the space of the described variable along which it reduces the error is $F\Sigma_T b$.

Theorem 11. (a) A one-dimensional description meets the distortion constraint with finite leakage if and only if $\Delta < \lambda_{\text{max}}(W^{1/2}\Sigma_Y W^{1/2})$, where $\Sigma_Y = F\Sigma_T F^\top$, and the least leakage among such descriptions is

$$L_{1d}(D) = \frac{1}{2} \log \left(1 + \frac{\Delta}{\mu_{\text{max}}(\Delta)} \right), \quad (8)$$

where $\mu_{\text{max}}(\Delta) > 0$ is the largest root of $\det(A(\Delta) - \mu \Sigma_{T|S}) = 0$. It is attained with b a corresponding generalized eigenvector and $\sigma^2 = b^\top \Sigma_T Q \Sigma_T b / \Delta - b^\top \Sigma_T b$, at which the distortion constraint is active.

(b) Suppose $\Delta < \lambda_{\text{max}}(W^{1/2}\Sigma_Y W^{1/2})$ and that $\mu_{\text{max}}(\Delta)$ is simple (if it is not, two distinct one-dimensional optima exist, their midpoint is strictly better by Theorem 5, and $L(D) < L_{1d}(D)$). Let Σ_{e0} and Σ_e be the error covariances of the description of (a), and let $\Gamma(\varpi) = \frac{1}{2}\Sigma_e - \varpi \Sigma_{e0} Q \Sigma_{e0}$. Then $L(D) = L_{1d}(D)$ if and only if there is $\varpi \geq 0$ with $\Gamma(\varpi) \succeq 0$ and $\Gamma(\varpi)b = 0$, and then $\varpi = \frac{1}{2}b^\top \Sigma_e b / (b^\top \Sigma_{e0} Q \Sigma_{e0} b)$.

(c) If the reproduction is scalar ($m = 1$), then $L(D) = L_{1d}(D)$ for every $0 < D < D_{\text{max}}$.

Proof. (a) For the description Z , $\text{Cov}(T | Z) = \Sigma_T - \Sigma_T b b^\top \Sigma_T / (b^\top \Sigma_T b + \sigma^2)$, so the distortion of $\mathbb{E}[Y | Z]$ is $D_{\text{max}} - b^\top \Sigma_T Q \Sigma_T b / (b^\top \Sigma_T b + \sigma^2)$, and the constraint reads $\sigma^2 \leq b^\top \Sigma_T Q \Sigma_T b / \Delta - b^\top \Sigma_T b$. The leakage $I(T; Z | S) = \frac{1}{2} \log(1 + b^\top \Sigma_{T|S} b / \sigma^2)$ decreases in σ^2 , so at the largest admissible σ^2 , where the constraint is active, it equals $\frac{1}{2} \log(1 + \Delta b^\top \Sigma_{T|S} b / b^\top A(\Delta) b)$, finite exactly when $b^\top A(\Delta) b > 0$. Minimizing over b is maximizing the Rayleigh quotient $b^\top A(\Delta) b / b^\top \Sigma_{T|S} b$ of a symmetric pencil with $\Sigma_{T|S} \succ 0$, whose maximum is μ_{max} . Some b has $b^\top A(\Delta) b > 0$ if and only if $\Sigma_T^{1/2} Q \Sigma_T^{1/2} \not\preceq \Delta I$, and the nonzero eigenvalues of that matrix are those of $W^{1/2}\Sigma_Y W^{1/2}$.

(b) The description of (a) has an active constraint and $\Sigma_e^{-1} = \Sigma_{T|S}^{-1} + bb^\top/\sigma^2$, so $\Sigma_{T|S} - \Sigma_e$ is a positive multiple of $\Sigma_{T|S}bb^\top\Sigma_{T|S}$ and $\Sigma_e^{-1}\Sigma_{T|S}b$ is a positive multiple of b . With $\Theta = \Sigma_e^{-1}\Gamma(\varpi)\Sigma_e^{-1}$, the conditions (4) at this point read $\Gamma(\varpi) \succeq 0$ and $\Gamma(\varpi)b = 0$, and Theorem 5(c) makes them necessary and sufficient. Taking $b^\top\Gamma(\varpi)b = 0$ gives ϖ .

(c) By Lemma 1 the optimum is attained by a jointly Gaussian channel $\hat{Y} = a^\top T + N$. If $\text{Cov}(Y, \hat{Y}) = 0$ its distortion is at least $D_{\max} > D$; otherwise $\mathbb{E}[Y | \hat{Y}]$ is a nonzero multiple of $\hat{Y}$, which leaves the leakage unchanged and does not raise the distortion, so the optimum is a one-dimensional description. $\square$

Corollary 12. *Let $T = (Y, V)$ be scalar with unit variances and correlation ρ , $|\rho| < 1$, let $W = 1$, and let $S = V + U$ with $\text{Var } U = \tau^2 > 0$; write $s = 1 + \tau^2$. For $0 < D < 1$,*

$$L(D) = \frac{1}{2} \log g^*, \quad g^* = \frac{(D + s - \rho^2) + \sqrt{(D + s - \rho^2)^2 - 4Ds(1 - \rho^2)}}{2Ds},$$

the larger root of $P(g) = Dsg^2 - (D + s - \rho^2)g + (1 - \rho^2)$, attained by $\hat{Y} = aY + bV + N$ with $a = (g^ - 1)/g^*$, $b = (g^* - 1)\rho/(g^*(g^*s - 1))$, and the noise variance that makes the distortion equal to D .*

Proof. By Theorem 11(c), $L(D) = \frac{1}{2} \log(1 + (1 - D)/\mu_{\max})$ with $\Delta = 1 - D$. Substituting $\mu = (1 - D)/(g - 1)$ into $\det(A(1 - D) - \mu\Sigma_{T|S})$, with $\Sigma_T = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$, $Q = e_1e_1^\top$, and $J = e_2e_2^\top/\tau^2$, gives $-(1 - D)(1 - \rho^2)P(g)/(s(g - 1)^2)$, a direct computation confirmed by computer algebra (Appendix C). Since $P(1) = (D - 1)\tau^2 < 0$ and $P(0) = 1 - \rho^2 > 0$, P has one root in $(0, 1)$, which corresponds to a negative μ , and one root $g^* > 1$, which corresponds to the unique positive root $\mu_{\max}$. The read direction is the corresponding generalized eigenvector; normalizing the reproduction gives the stated coefficients. $\square$

For a scalar described variable with vector context, Theorem 11(c) likewise gives $L(D)$ as an explicit function of the largest generalized eigenvalue of a $p \times p$ pencil, $p = 1 + \dim V$.

VI. DIRECTIONS AND MISALIGNMENT

Proposition 13. *With no holder ($J = 0$), the best one-dimensional description reads the top generalized eigenvector of the pencil $(\Sigma_T Q \Sigma_T, \Sigma_T)$ at every budget, and the rate-optimal description reverse water-fills on the eigenvalues of $W^{1/2} \Sigma_Y W^{1/2}$ along fixed directions.*

Proof. With $J = 0$, $\Sigma_{T|S} = \Sigma_T$ and $A(\Delta)b = \mu\Sigma_T b$ reads $\Sigma_T Q \Sigma_T b = (\mu + \Delta)\Sigma_T b$, whose eigenvectors do not depend on Δ . The rate-optimal description is the weighted quadratic rate–distortion solution [15]: in the coordinates $W^{1/2}Y$ the distortion is squared error, and reverse water-filling describes fixed principal components [3]. $\square$

The Gaussian information bottleneck has the same property: its optimal projections are fixed eigenvectors, and its tradeoff parameter sets only their number and scale [4]. The leakage-optimal description behaves differently.

Theorem 14. *Let $\mu_{\max}(\Delta)$ be simple and positive on an open interval $\mathcal{I} \subseteq (0, \lambda_{\max}(W^{1/2} \Sigma_Y W^{1/2}))$, with generalized eigenvector $b(\Delta)$. Then $b(\Delta)$ spans the same line for every $\Delta \in \mathcal{I}$ if and only if, for some $\Delta_0 \in \mathcal{I}$, $b_0 = b(\Delta_0)$ satisfies*

$$Q\Sigma_T b_0 = \lambda b_0 \quad \text{and} \quad J\Sigma_T b_0 = \kappa b_0 \tag{9}$$

for scalars $\lambda > 0$ and $\kappa \geq 0$, that is, if and only if b_0 is a common generalized eigenvector of the pencils $(\Sigma_T Q \Sigma_T, \Sigma_T)$ and $(\Sigma_T J \Sigma_T, \Sigma_T)$. Otherwise $b(\Delta)$ is constant on no open subinterval of $\mathcal{I}$.

Proof. A simple eigenvalue of a symmetric pencil with positive definite second matrix, and its eigenvector, are analytic in the parameter [21]. Write $A(\Delta) = A_0 - \Delta\Sigma_T$.

Suppose b is constant on an open subinterval. Differentiating $(A_0 - \Delta\Sigma_T)b = \mu_{\max}(\Delta)\Sigma_{T|S}b$ gives $\Sigma_T b = c\Sigma_{T|S}b$ for a scalar c , and $c = b^\top \Sigma_T b / b^\top \Sigma_{T|S} b \geq 1$ because $\Sigma_{T|S} \preceq \Sigma_T$. Then $A_0 b = (\Delta + \mu_{\max}/c)\Sigma_T b$, that is, $Q\Sigma_T b = \lambda b$ with $\lambda = \Delta + \mu_{\max}/c > 0$; and multiplying $\Sigma_T b = c\Sigma_{T|S}b$ by $\Sigma_{T|S}^{-1} = \Sigma_T^{-1} + J$ gives $J\Sigma_T b = (c - 1)b$, so $\kappa = c - 1 \geq 0$. This proves the last sentence and the forward direction.

Conversely, if b_0 satisfies (9), then $\Sigma_{T|S}^{-1}\Sigma_T b_0 = (1 + \kappa)b_0$, so $\Sigma_T b_0 = (1 + \kappa)\Sigma_{T|S}b_0$ and $A(\Delta)b_0 = (\lambda - \Delta)\Sigma_T b_0 = (\lambda - \Delta)(1 + \kappa)\Sigma_{T|S}b_0$ for every Δ : b_0 is a generalized eigenvector with eigenvalue $(\lambda - \Delta)(1 + \kappa)$ throughout. Let $\mathcal{E} = \{\Delta \in \mathcal{I} : (\lambda - \Delta)(1 + \kappa) = \mu_{\max}(\Delta)\}$. It contains Δ_0 and is closed in $\mathcal{I}$ by continuity. It is

open, because near a point of $\mathcal{E}$ the simple eigenvalue $\mu_{\max}$ is the only eigenvalue of the pencil close to it, and $(\lambda - \Delta)(1 + \kappa)$ is an eigenvalue close to it. Since $\mathcal{I}$ is connected, $\mathcal{E} = \mathcal{I}$, and by simplicity $b(\Delta)$ spans the line of b_0 throughout. $\square$

Corollary 15. *Suppose the holder observes only the context: in coordinates $T = (Y, V)$, $F = [I \ 0]$ and $H = [0 \ H_V]$. Then (9) holds if and only if $b_0 = F^\top \beta$ with $W\Sigma_Y \beta = \lambda \beta$ and $\text{Cov}(S, \beta^\top Y) = 0$.*

Proof. $Q\Sigma_T b_0 = \lambda b_0$ with $\lambda > 0$ puts b_0 in the range of Q , which is the range of $F^\top$, so $b_0 = F^\top \beta$ and $F^\top W\Sigma_Y \beta = \lambda F^\top \beta$. The vector $J\Sigma_T b_0$ lies in the range of $H^\top$, which meets the range of $F^\top$ only at zero, so $J\Sigma_T b_0 = \kappa b_0$ forces $J\Sigma_T b_0 = 0$; with $w = \Sigma_T b_0$, $w^\top Jw = 0$ gives $\Sigma_U^{-1/2} Hw = 0$, that is, $\Sigma_U^{-1} H\Sigma_T F^\top \beta = 0$, which is $\text{Cov}(S, \beta^\top Y) = 0$. The converse is direct. $\square$

The same condition decides whether the two optimal descriptions coincide. Let b_R be the top generalized eigenvector of $(\Sigma_T Q \Sigma_T, \Sigma_T)$, the read direction of the best one-dimensional rate description.

Theorem 16. *Let $\lambda_1 \geq \lambda_2$ be the two largest eigenvalues of $W^{1/2} \Sigma_Y W^{1/2}$ and suppose $\Delta \leq \lambda_1 - \lambda_2$, the regime in which the rate-optimal description is one-dimensional, $b_R^\top T + N$. If $J\Sigma_T b_R$ is not parallel to b_R , the rate-optimal description is not leakage-optimal and the leakage-optimal description is not rate-optimal, so each pays a strictly positive excess in the other cost. If $J\Sigma_T b_R$ is parallel to b_R , then b_R is a generalized eigenvector of the leakage pencil at every Δ , and inside the one-dimensional regime of Theorem 11 the two descriptions coincide exactly when it is the top one.*

Proof. In the stated regime reverse water-filling describes only the principal component, so the rate-optimal description is the one-dimensional description reading b_R , where $Q\Sigma_T b_R = \lambda_1 b_R$. If it were leakage-optimal, then $L(D) = L_{1d}(D)$ and b_R would maximize the Rayleigh quotient of Theorem 11(a), hence be a generalized eigenvector of $(A(\Delta), \Sigma_{T|S})$, that is, $(\lambda_1 - \Delta)\Sigma_T b_R \propto \Sigma_{T|S} b_R$; multiplying by $\Sigma_{T|S}^{-1} = \Sigma_T^{-1} + J$ gives $J\Sigma_T b_R \propto b_R$. So when $J\Sigma_T b_R$ is not parallel to b_R the rate-optimal description is not leakage-optimal, and since both minimizers of Theorem 5(a) are unique, the leakage-optimal description is not rate-optimal either. The same computation read backward shows that $J\Sigma_T b_R \propto b_R$ makes b_R a generalized eigenvector at every Δ , and inside the one-dimensional regime the leakage-optimal description reads the top one. $\square$

In the scalar case of Corollary 12, $b_R = e_1$ and $J\Sigma_T e_1 = (\rho/\tau^2)e_2$, which is parallel to e_1 only when $\rho = 0$. So the two descriptions differ exactly when the context is correlated with the described variable. The rate-optimal channel is the classical reverse channel, with leakage $\frac{1}{2} \log[((1-D)(1-\rho^2/s) + D)/D]$, strictly above $L(D)$, and the leakage-optimal channel of Corollary 12 has rate strictly above $\frac{1}{2} \log(1/D)$. The rate excess is unbounded toward the clean-context boundary $\tau^2 \rightarrow 0$ at $D = 1 - \rho^2$, where the leakage-optimal channel pins the reproduction to the context.

Corollary 17. *There are sources for which the leakage-optimal one-dimensional read takes different directions at two budgets. For such a source no description problem for a single party with a fixed weighted quadratic distortion has the leakage-optimal descriptions as its optimal descriptions at both budgets.*

Proof. Take the scalar source of Corollary 12 with $\rho \neq 0$. By Theorem 11(c) the leakage-optimal description is the one-dimensional description at every D , and its read direction is proportional to $(g^* s - 1, \rho)$, the ratio of the coefficients in Corollary 12. Since ℓ is strictly decreasing (Lemma 20), g^* is strictly decreasing in D , so the read line differs at any two budgets. By Proposition 13, applied to any fixed weight on T , the best one-dimensional description for a single party reads the same line at both budgets. The coupled instance of Section IX, whose leakage-optimal descriptions reduce the error along 47.48° and 41.15° at two budgets, illustrates the same effect in the space of the described variable. $\square$

A description that reads a fixed direction, such as one designed for a single party, therefore loses something at all but isolated budgets. The loss has an explicit lower bound. Let $\mu_1 \geq \mu_2 \geq \dots$ be the roots of $\det(A(\Delta) - \mu \Sigma_{T|S}) = 0$, and measure angles between directions in the inner product of $\Sigma_{T|S}$, which does not depend on the budget.

Proposition 18. Let $0 < \Delta < \lambda_{\max}(W^{1/2}\Sigma_Y W^{1/2})$ and let $b(\Delta)$ be a generalized eigenvector for $\mu_1 > 0$. For b with $\mu_b = b^\top A(\Delta)b/b^\top \Sigma_{T|S} b > 0$, the one-dimensional description reading b at the active noise level has leakage $\ell_b(\Delta) = \frac{1}{2} \log(1 + \Delta/\mu_b)$, and

$$\ell_b(\Delta) - L_{1d}(D) \geq \frac{1}{2} \log\left(1 + \frac{\Delta(\mu_1 - \mu_2) \sin^2 \theta}{\mu_1(\mu_1 + \Delta)}\right), \quad (10)$$

where θ is the angle between b and $b(\Delta)$.

Proof. The leakage is the one computed in the proof of Theorem 11(a). In a $\Sigma_{T|S}$ -orthonormal basis of generalized eigenvectors v_i , with $v_1 = b(\Delta)/\|b(\Delta)\|_{\Sigma_{T|S}}$, write $b = \sum_i c_i v_i$. Then $\mu_b = \sum_i \mu_i c_i^2 / \sum_i c_i^2$, so $\mu_1 - \mu_b \geq (\mu_1 - \mu_2) \sum_{i \geq 2} c_i^2 / \sum_i c_i^2 = (\mu_1 - \mu_2) \sin^2 \theta$. Finally $(1 + \Delta/\mu_b)/(1 + \Delta/\mu_1) = 1 + \Delta(\mu_1 - \mu_b)/(\mu_b(\mu_1 + \Delta))$, and $\mu_b \leq \mu_1$. $\square$

Theorem 19. Let Δ_1 and Δ_2 satisfy the hypotheses of Proposition 18, and let θ_{12} be the angle between $b(\Delta_1)$ and $b(\Delta_2)$. Every one-dimensional description that reads the same direction b at both budgets has, at one of them, excess leakage over L_{1d} at least

$$\min_{i=1,2} \frac{1}{2} \log\left(1 + \frac{\Delta_i(\mu_1 - \mu_2) \sin^2(\theta_{12}/2)}{\mu_1(\mu_1 + \Delta_i)}\right),$$

with μ_1 and μ_2 evaluated at Δ_i . By Theorem 14 and analyticity, unless the read is constant, for each Δ_1 the angle θ_{12} is positive for all but isolated Δ_2 .

Proof. The angle between lines is a metric on the projective space of the $\Sigma_{T|S}$ inner product, so the angles from b to $b(\Delta_1)$ and to $b(\Delta_2)$ sum to at least θ_{12} , and one of them is at least $\theta_{12}/2$. The bound (10) increases with $\sin^2 \theta$ on $[0, \pi/2]$. $\square$

Where $L(D) = L_{1d}(D)$, which holds at every budget for a scalar described variable and inside the certificate of Theorem 11(b) in general, these are excesses over the least leakage of any description. The bounds need no search and are conservative; the minimax values below were found by search over directions. In the coupled instance the rate read pays 0.139 bits of excess leakage at $D = 1.1$ (bound 0.045) and 0.011 bits at $D = 1.38$ (bound 0.0016). Yet the best fixed read for the two budgets $D = 1.38$ and 1.1, whose optimal reads are 24.8° apart, pays only 0.0003 bits at the worse of them: there the price is reading the wrong direction, not failing to turn. A cleaner context makes turning costly. For the scalar source with $(\rho^2, \tau^2) = (0.75, 0.05)$ and budgets $D = 0.1$ and 0.7, the optimal reads are 49.8° apart and every fixed read pays at least 0.0245 bits at one budget (bound 0.0039), more than the least leakage 0.0216 bits at $D = 0.7$. With $(\rho^2, \tau^2) = (0.5, 10^{-3})$ and $D = 0.2$ and 0.8, the reads are 84.0° apart and every fixed read pays at least 0.0386 bits, against a least leakage of 0.0005 bits at $D = 0.8$.

VII. INDEPENDENT COMPONENTS

Suppose $Y = (Y_1, \dots, Y_m)$, W is diagonal with entries w_j , and each Y_j has its own context V_j and holder observation $S_j = V_j + U_j$, the triples (Y_j, V_j, S_j) being independent, with $\text{Var } Y_j = \sigma_j^2$, $\text{Var } V_j = 1$, $\mathbb{E}[Y_j V_j] = \rho_j \sigma_j$, $|\rho_j| < 1$, and $\text{Var } U_j = \tau_j^2 > 0$. Absorb w_j into σ_j^2 . Write ℓ_j for the scalar function $\delta \mapsto \frac{1}{2} \log g^*(\delta)$ of Corollary 12 at (ρ_j, τ_j^2) , extended by zero for $\delta \geq 1$, and $L_j(D_j) = \ell_j(D_j/\sigma_j^2)$.

Lemma 20. For $0 < \delta < 1$, with $g = g^*(\delta)$, $s = 1 + \tau^2$, $k = gs - 1$, and $\Delta_\delta = \sqrt{(\delta + s - \rho^2)^2 - 4\delta s(1 - \rho^2)} > 0$,

$$\ell'(\delta) = -\frac{k}{2 \ln 2 \Delta_\delta}, \quad \ell''(\delta) = \frac{(1 - \rho^2)k^3 + \rho^2 \tau^2 g^2 (\tau^4 g + \tau^2 (2g + 1) + g + 1)}{2 \ln 2 g \Delta_\delta^3} > 0,$$

so ℓ is strictly convex and decreasing, $\ell' \rightarrow -\infty$ as $\delta \rightarrow 0$, and $\ell' \rightarrow -\tau^2/(2 \ln 2(\rho^2 + \tau^2))$ as $\delta \rightarrow 1$.

Proof. The larger root is simple, with $\partial_\delta P = \Delta_\delta$ there, so g^* is smooth; differentiating $P(g^*(\delta)) = 0$ with $\partial_\delta P = gk$ gives $g' = -gk/\Delta_\delta$ and the first formula. Differentiating once more and eliminating δ through $\delta gk = (s - \rho^2)g - (1 - \rho^2)$ gives the second, an identity of polynomials confirmed by computer algebra (Appendix C); its numerator is positive. The limits follow from $g^* \rightarrow \infty$, $\Delta_\delta \rightarrow s - \rho^2$ as $\delta \rightarrow 0$ and $g^* \rightarrow 1$, $k \rightarrow \tau^2$, $\Delta_\delta \rightarrow \rho^2 + \tau^2$ as $\delta \rightarrow 1$. $\square$

Define the activation level $\kappa_j = -L'_j(\sigma_j^2 -) = \tau_j^2 / (2 \ln 2 \sigma_j^2 (\rho_j^2 + \tau_j^2))$; for the rate it would be $1/(2 \ln 2 \sigma_j^2)$. These components are two-dimensional blocks of Theorem 7, each solved by Corollary 12.

Theorem 21. (a) $\mathcal{R}(D)$ is the union, over $D_j \geq 0$ with $\sum_j D_j \leq D$, of the Minkowski sums of the scalar regions of the components, attained by product channels. (b) $L(D) = \min \sum_j L_j(D_j)$ over such allocations, with a unique minimizer $D_j = D_j(\theta^*)$, where $D_j(\theta) = \sigma_j^2$ for $\theta \leq \kappa_j$ and otherwise solves $-L'_j(D_j) = \theta$, and θ^* is the unique level with $\sum_j D_j(\theta^*) = D$. (c) As D decreases, the leakage-optimal description begins to describe component j in decreasing order of $\sigma_j^2(1 + \rho_j^2/\tau_j^2)$, while the rate-optimal description proceeds in decreasing order of σ_j^2 .

Proof. (a) By Theorem 3 it suffices to identify the single-letter union. For an admissible channel with $D_j = \mathbb{E}(Y_j - \hat{Y}_j)^2$, each induced channel $p(\hat{y}_j | t_j)$ satisfies $\hat{Y}_j - T_j - S_j$, and since the pairs (T_j, S_j) are independent the chain rule gives $I(T; \hat{\mathbf{Y}}) \geq \sum_j I(T_j; \hat{Y}_j)$ and $I(T; \hat{\mathbf{Y}} | S) = \sum_j [I(T_j; \hat{\mathbf{Y}}, S, T^{j-1}) - I(T_j; S_j)] \geq \sum_j I(T_j; \hat{Y}_j | S_j)$. A product of component channels attains the sum with equality. (b) Each L_j is convex and nonincreasing, strictly convex on $(0, \sigma_j^2)$ by Lemma 20, with left derivative $-\kappa_j$ at σ_j^2 . An allocation is optimal if and only if some $\theta \geq 0$ lies in $-\partial L_j(D_j)$ for every j with $\theta(\sum_j D_j - D) = 0$. Since $D < \sum_j \sigma_j^2$, $\theta > 0$; no D_j is 0 because $-L'_j \rightarrow \infty$ there; and $\sum_j D_j(\theta)$ decreases continuously and strictly from $\sum_j \sigma_j^2$ at $\min_j \kappa_j$ to 0, so θ^* is unique. (c) $D_j(\theta) < \sigma_j^2$ exactly when $\theta > \kappa_j$, and θ^* increases as D decreases. $\square$

Corollary 22. If $m = 2$, $\sigma_1^2 > \sigma_2^2$, and $\sigma_1^2(1 + \rho_1^2/\tau_1^2) < \sigma_2^2(1 + \rho_2^2/\tau_2^2)$, then for every $D \in [\max\{2\sigma_2^2, \sigma_1^2 + D_2(\kappa_1)\}, \sigma_1^2 + \sigma_2^2]$ the rate-optimal description describes component 1 alone and the leakage-optimal description describes component 2 alone.

Proof. Reverse water-filling describes component 1 alone exactly for $2\sigma_2^2 \leq D < \sigma_1^2 + \sigma_2^2$; Theorem 21(b) describes component 2 alone exactly when $\kappa_2 < \theta^* \leq \kappa_1$, that is, for $\sigma_1^2 + D_2(\kappa_1) \leq D < \sigma_1^2 + \sigma_2^2$. $\square$

VIII. DECODER SIDE INFORMATION

Suppose the decoder also observes $C = H_C T + N_C$, with $N_C \sim \mathcal{N}(0, \Sigma_{N_C})$, $\Sigma_{N_C} \succ 0$, independent of everything else, and reconstructs from (M, C^n) . Write $J_C := H_C^T \Sigma_{N_C}^{-1} H_C$ and $\Sigma_{T|C} = (\Sigma_T^{-1} + J_C)^{-1}$. The single-letter region is Villard and Piantanida's [1, Theorem 3] with uncoded side information at the decoder: $R \geq I(V; T | C)$, distortion at most $\mathbb{E}d(Y, \hat{\mathbf{Y}}(V, C))$, and equivocation at most $h(T | V, C) + I(T; C | U) - I(T; S | U)$, over $U - V - T - (C, S)$. We use it for the Gaussian source as Ekrem and Ulukus do [2, Sec. 2], without reproving its extension to continuous alphabets. The two auxiliaries are what makes the general case hard; Ekrem and Ulukus evaluate the two endpoints separately, and a joint characterization is reported by Xu and Chen [5]. The two degraded cases are explicit and inherit the structure of the preceding sections.

For Gaussian linear observations the information matrices order the side informations: if $J_S \preceq J_C$, the holder's observation is a stochastically degraded version of the decoder's. Indeed, the sufficient statistic $J_C T + n_C$, $n_C \sim \mathcal{N}(0, J_C)$, of C is mapped by $A = J_S J_C^+$ to $J_S T + A n_C$, and $A J_C A^T = J_S^{1/2} (J_S^{1/2} J_C^+ J_S^{1/2}) J_S^{1/2} \preceq J_S$, so adding independent noise of covariance $J_S - A J_C A^T$ reproduces the law of the sufficient statistic $J_S T + n_S$ of S . Degradedness implies that the decoder's side information is less noisy than the holder's in the sense of [1], and symmetrically when $J_C \preceq J_S$.

Lemma 23. Let $V - T - (C, S)$ with $J_C \preceq J_S$, and let $M_C = \mathbb{E}[\text{Cov}(T | V, C)]$ and $M_S = \mathbb{E}[\text{Cov}(T | V, S)]$ be the minimum mean-square error covariances. Then $M_S \preceq (M_C^{-1} + J_S - J_C)^{-1}$, with equality when (V, T) is jointly Gaussian.

Proof. The conditional law of T given (V, S) depends on S only through its sufficient statistic, whose conditional law given T is that of the sum of the sufficient statistics of two independent observations of T with information matrices J_C and $\Delta J = J_S - J_C$. So M_S is the error covariance of estimating T from V , an observation C' equivalent to C , and an independent observation Z with information ΔJ , and (T, V, C') has the law of (T, V, C) . For each realization w of (V, C') , the conditional covariance of T is C_w , and the best linear estimate of T from Z under that conditional law has error covariance $(C_w^{-1} + \Delta J)^{-1}$, which bounds the conditional minimum mean-square error. Averaging over w and using the matrix concavity of $X \mapsto (X^{-1} + \Delta J)^{-1}$ (proof of Theorem 5), the operator Jensen inequality gives $M_S \preceq \mathbb{E}[(C_w^{-1} + \Delta J)^{-1}] \preceq (M_C^{-1} + \Delta J)^{-1}$. When (V, T) is jointly Gaussian every inequality is an equality, because C_w is constant and the estimate is linear. $\square$

Theorem 24. *Under the single-letter region above, for $0 < D < \text{tr}(Q\Sigma_{T|C})$:*

- (a) *if $J_S \preceq J_C$, the region is the quadrant above $(R_C(D), R_C(D))$, where $R_C(D) = \min\{\frac{1}{2}\log(\det \Sigma_{T|C}/\det M) : 0 \prec M \preceq \Sigma_{T|C}, \text{tr}(QM) \leq D\}$ is reverse water-filling on the eigenvalues of $\Sigma_{T|C}^{1/2}Q\Sigma_{T|C}^{1/2}$; the leakage-optimal and rate-optimal descriptions coincide, and the holder learns as much as the rate;*
- (b) *if $J_C \preceq J_S$, the region is the region of Proposition 2 for the source $\mathcal{N}(0, \Sigma_{T|C})$ with holder information $J_S - J_C$ and the same weight Q ; the coordinate is the decoder's error covariance $M = \text{Cov}(T | V, C)$, and Theorems 5, 7, 8, 11, 14, and 16 hold with Σ_T replaced by $\Sigma_{T|C}$ and J by $J_S - J_C$.*

Proof. For any V , the divergence bound in the proof of Lemma 1, stopped before linear estimation enters, gives $I(T; V | C) \geq \frac{1}{2}\log(\det \Sigma_{T|C}/\det M_C)$ and $I(T; V | S) \geq \frac{1}{2}\log(\det \Sigma_{T|S}/\det M_S)$; and the distortion of any reproduction from (V, C) is at least $\text{tr}(QM_C)$.

(a) The decoder's side information is less noisy, so U constant is optimal [1, Cor. 2], and with $L = h(T | S) - \Delta$ the leakage bound becomes $L \geq h(T | S) - h(T | V, C) - I(T; C) + I(T; S) = I(T; V | C)$, the same as the rate bound. The union over V of the quadrants above $(I(T; V | C), I(T; V | C))$ is the quadrant above $(R_C(D), R_C(D))$, where $R_C(D)$ is the least $I(T; V | C)$ at distortion D , the Gaussian weighted Wyner–Ziv function; the bounds above with $M = M_C$ and a jointly Gaussian V with $\text{Cov}(T | V, C) = M$ give its value.

(b) The holder's side information is less noisy, so $U = V$ is optimal [1, Cor. 3], the equivocation bound is $h(T | V, S)$, and $L \geq I(T; V | S)$. By the bounds above and Lemma 23, $R \geq \frac{1}{2}\log(\det \Sigma_{T|C}/\det M_C)$, $L \geq \frac{1}{2}\log \det \Sigma_{T|S} + \frac{1}{2}\log \det(M_C^{-1} + J_S - J_C)$, and the distortion is at least $\text{tr}(QM_C)$. A jointly Gaussian V with precision $M^{-1} - \Sigma_{T|C}^{-1} \succeq 0$ has $M_C = M$ and attains all three. Since $(\Sigma_{T|C}^{-1} + J_S - J_C)^{-1} = \Sigma_{T|S}$, these are the coordinates (1) of the source $\mathcal{N}(0, \Sigma_{T|C})$ with holder information $J_S - J_C$ at $\Sigma_{e0} = M$, over the same feasible set. $\square$

So a decoder that knows more than the holder removes the tradeoff, and a holder that knows more than the decoder leaves exactly the problem of the preceding sections for the source as the decoder already knows it, with the holder's excess information in place of its total. When neither dominates, the auxiliary U can matter, and the region is beyond the methods of this paper.

IX. INSTANCES

Scalar. At $(\rho^2, \tau^2, D) = (0.75, 0.5, 0.3)$ the rate-optimal channel pays 0.0349 bits of excess leakage and the leakage-optimal channel 0.0400 bits of excess rate; over the box $\rho^2 \in [0.05, 0.95]$, $\tau^2 \in [\frac{1}{4}, 4]$, $D \in [0.05, 0.95]$ the rate excess of the leakage-optimal channel stays below 0.114 bits and the leakage excess of the rate-optimal channel below 0.077 bits, and at $(\rho^2, \tau^2, D) = (0.5, 10^{-3}, 0.5)$ the rate excess reaches 1.537 bits. Two sources with the same law of (Y, S) up to scaling of S , $(\rho^2, \tau^2) = (\frac{3}{5}, \frac{1}{5})$ and $(\frac{3}{4}, \frac{1}{2})$, have $L = 1.1880$ and 1.2105 bits at $D = 0.1$: the leakage is not determined by the law of (Y, S) , because the encoder observes the context.

Coupled. Take $\Sigma_Y = \text{diag}(1, 0.4)$, $W = I$, a scalar context with $\text{Var } V = 1$ and $\text{Cov}(Y, V) = 0.6u$ for the unit vector u at 50° , and $S = V + U$ with $\text{Var } U = 0.05$. The certificate of Theorem 11(b) holds for $D \geq 1.08530$, out of $D_{\max} = 1.4$. The rate-optimal description reduces the error along the first axis for every $D \geq 0.8$. The leakage-optimal description reduces it along 47.48° , 46.76° , 45.07° , and 41.15° at $D = 1.38, 1.3, 1.2$, and 1.1 . Four sources with the same u and the same law of (Y, S) up to scaling, $(|\text{Cov}(Y, V)|^2, \tau^2) = (0.33, 0.1)$, $(0.36, 0.2)$, $(0.39, 0.3)$, $(0.42, 0.4)$, read along 42.83° , 37.61° , 33.75° , and 30.84° at $D = 1.3$, each inside its certificate regime: the direction, too, is not determined by the law of (Y, S) .

Independent components. With $(\sigma_1^2, \rho_1^2, \tau_1^2) = (1, 0, 1)$ and $(\sigma_2^2, \rho_2^2, \tau_2^2) = (0.6, 0.9, 0.05)$, $\kappa_1 = 0.7213$, $\kappa_2 = 0.0633$, and the two descriptions describe disjoint components for every $D \in [1.2002, 1.6)$. At $D = 1.4$ the rate-optimal description pays $R = L = 0.1610$ bits on component 1 and the leakage-optimal description pays $L = 0.0197$ and $R = 0.3272$ bits on component 2.

X. PRACTICAL QUANTIZERS

The theorems concern long blocks and Gaussian test channels. A practical description is often a fixed-rate scalar quantizer applied to one projection of the data. We ask two questions of such a quantizer. Does the projection it reads change its leakage at equal rate and equal distortion? And does the leakage-aware projection turn with the budget, as Theorem 14 predicts for the optimum?

TABLE I

LLOYD-MAX QUANTIZERS ON ONE PROJECTION OF THE COUPLED INSTANCE. AT EACH BUDGET THE LEAKAGE-AWARE PROJECTION LEAKS LESS THAN THE PRINCIPAL AXIS AT THE SAME RATE, AND IT READS A DIRECTION THAT TURNS FURTHER THAN THE GAUSSIAN OPTIMUM. LEAKAGE IN BITS, ANGLES IN DEGREES IN THE PLANE OF Y .

k	$H(M)$	f	D_0	principal	aware	reduction	angle (aware / pencil)	$L_{1d}(D_0)$
2	1.9111	0.15	0.6499	1.8147	1.4205	21.7%	16.40 / 6.83	0.8758
		0.40	0.8705	1.8147	0.8956	50.6%	32.09 / 20.04	0.3913
		0.70	1.1352	1.8147	0.5066	72.1%	49.93 / 42.94	0.0636
3	2.8249	0.15	0.5794	2.7188	2.2667	16.6%	16.50 / 4.36	1.1201
		0.40	0.8207	2.7188	1.5896	41.5%	31.78 / 16.09	0.4803
		0.70	1.1104	2.7188	0.9714	64.3%	50.04 / 41.73	0.0792
4	3.7653	0.15	0.5581	3.6563	3.1855	12.9%	16.58 / 3.71	1.2126
		0.40	0.8057	3.6563	2.4463	33.1%	32.02 / 15.02	0.5088
		0.70	1.1029	3.6563	1.6851	53.9%	50.01 / 41.32	0.0845

Setup. Take the coupled instance of Section IX, so that $Q = F^\top F$ and $D_{\max} = \text{tr } \Sigma_Y = 1.4$. For a direction b let $Z = b^\top T / (b^\top \Sigma_T b)^{1/2}$, which is standard normal, and let $M = q(Z)$ for a scalar quantizer q with K levels and thresholds at the midpoints between levels. The decoder reproduces $\hat{Y} = \mathbb{E}[Y | M]$. Write p_i for the probability of cell i , m_i for $\mathbb{E}[Z | M = i]$, and $\gamma = \sum_i p_i m_i^2$. Let $c = \mathbb{E}[YZ] = F \Sigma_T b / (b^\top \Sigma_T b)^{1/2}$ and let ρ be the correlation of Z with the scalar S . Because $Y - cZ$ is independent of Z , the distortion is

$$D(b) = \text{tr } \Sigma_Y - \gamma \|c\|^2.$$

The entropy $H(M) = -\sum_i p_i \log p_i$ does not depend on b , so every comparison below is at equal rate, whether the index is sent at $\log K$ bits or entropy coded. Given S , the variable Z is normal with variance $1 - \rho^2$ and a mean proportional to ρ , so the leakage $H(M | S)$ depends on b only through ρ .

Proposition 25. *For every quantizer q , the leakage $H(q(Z) | S)$ is a nonincreasing function of $|\rho|$.*

Proof. Let $\bar{S}$ be S scaled to unit variance. The law of $(Z, \bar{S})$ is fixed by ρ , and replacing $\bar{S}$ by $-\bar{S}$ flips the sign of ρ without changing $H(M | \bar{S})$, so the leakage is a function of $|\rho|$. For $0 \leq \rho < \rho' \leq 1$, let $(Z, \bar{S}')$ have correlation ρ' and let $\bar{S}'' = (\rho/\rho')\bar{S}' + (1 - \rho^2/\rho'^2)^{1/2}N$ with N standard normal and independent. Then $(Z, \bar{S}'')$ has correlation ρ and $Z - \bar{S}' - \bar{S}''$ is a Markov chain, so $H(M | \bar{S}'') \geq H(M | \bar{S}', \bar{S}'') = H(M | \bar{S}')$. $\square$

For every quantizer used here the leakage is strictly decreasing in $|\rho|$, which we checked at 200 values of $|\rho|$ in $[0, 0.995]$. The leakage-aware projection for a given quantizer therefore maximizes $|\rho|$ subject to $\gamma \|c\|^2 \geq \text{tr } \Sigma_Y - D_0$. We place each budget relative to the principal-axis quantizer, which reads Y_1 , has $c = (1, 0)$, and has distortion $D_R = \text{tr } \Sigma_Y - \gamma$. Setting $D_0 = D_R + f(\text{tr } \Sigma_Y - D_R)$ with $0 < f < 1$, the constraint becomes $\|c\|^2 \geq 1 - f$, and the quantizer has dropped out. At a given f the leakage-aware projection is the same for every quantizer, and only its leakage depends on the quantizer. For $f \geq 0.64$ it is the context itself. The choice $Z = V$ has $c = \text{Cov}(Y, V) = 0.6u$ with $\|c\|^2 = 0.36$, and V maximizes $|\rho|$ over all projections because $S = V + U$.

Computation. We use the Lloyd-Max quantizers for the standard normal law [26], [27] at $k = 2, 3$, and 4 bits, and $f = 0.15, 0.4$, and 0.7. The rate, the distortion, and the leakage are exact Gaussian integrals, the last by Gauss-Hermite quadrature over S . The leakage-aware projection is the direction of largest $|\rho|$ among 2×10^6 random feasible directions, and its costs are then evaluated exactly. Table I lists the results.

Results. At equal rate and equal budget, the leakage-aware quantizer leaks 12.9% to 72.1% less than the principal-axis quantizer (Table I and Fig. 1(a)). The gain grows as the budget loosens and is largest at 2 bits, where at $f = 0.7$ the quantizer leaks 0.5066 bits instead of 1.8147. The principal-axis quantizer reduces the error along the first axis at every budget. The leakage-aware quantizer reduces it along 16.4° to 16.6° , 31.8° to 32.1° , and 49.9° to 50.0° at the three values of f (Fig. 1(b)). The reduction above makes the direction the same for every quantizer, and the three agree within 0.31° , a spread left by the random search. They turn with the budget toward the context direction u at 50° , which the quantizer reaches once it reads V .

The Gaussian optimum of Theorem 11 turns the same way but less, through 3.7° to 6.8° , 15.0° to 20.0° , and 41.3° to 42.9° at the same budgets. A Gaussian description at budget D_0 chooses its noise variance as well as its direction, so it can spend part of the slack on coarser resolution. A fixed-rate quantizer has a fixed γ . It can

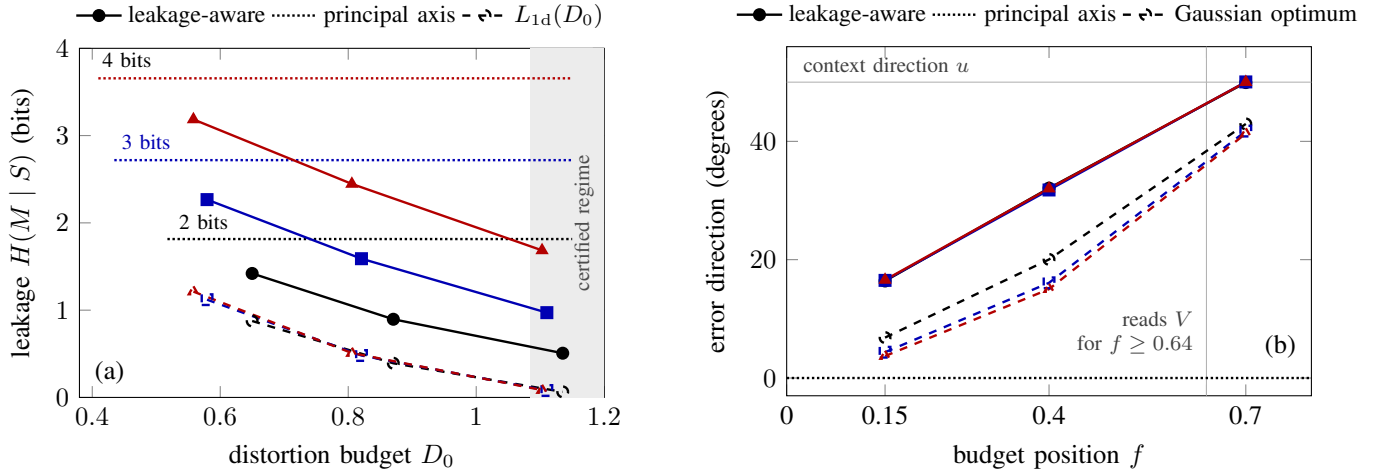


Fig. 1. Fixed-rate Lloyd–Max quantizers on one projection of the coupled instance, at 2 (circles), 3 (squares), and 4 bits (triangles). (a) Leakage at equal rate. The leakage-aware projection leaks less than the principal axis at every budget, and in the shaded certified regime $L_{1d}(D_0)$ is the least leakage of any description. (b) The direction in the plane of Y along which each description reduces the error. The leakage-aware quantizer turns toward the context direction and turns further than the Gaussian optimum, while the principal-axis quantizer stays at 0° .

spend the slack only on direction, and it turns further. Its leakage also stays well above the floor. Every budget at $f = 0.7$ lies in the certified regime $D \geq 1.08530$ of Section IX, where $L_{1d}(D_0) = L(D_0)$. Because $\hat{Y}$ is an admissible channel and $H(M | S) = I(T; M | S) \geq I(T; \hat{Y} | S)$, Theorem 3 makes $L(D_0)$ a lower bound on the leakage of the quantizer. The 2-bit quantizer leaks 0.5066 bits there, against a floor of 0.0636 bits. At $f = 0.15$ and 0.4 the budgets lie below the certified regime, and $L_{1d}(D_0)$ is the best value among one-dimensional Gaussian descriptions rather than a bound.

The 3-bit and 4-bit level tables shipped with one open-source vector-compression library differ from the Lloyd–Max levels. Their mean-square errors on the standard normal law are 0.0456 and 0.0108, against 0.0346 and 0.0095. With those tables the reductions range from 12.5% to 72.1%, and the directions agree with Table I within 0.31° .

XI. DISCUSSION

A description held by two parties has a cost for each, and the encoder’s observations decide how the costs trade off. For Gaussian sources without decoder side information the region is now explicit in structure: a convex program in the error covariance, a closed form in a one-dimensional regime, and directions that turn with the budget exactly when the two parties’ geometries are not aligned. Several problems remain open. With side information at the decoder, the degraded cases are settled here. When neither side information dominates, the second auxiliary can matter. For a matrix distortion on the whole source and a full-rank eavesdropper, Xu and Chen [5] show that Gaussian auxiliaries are optimal. Whether their two-precision program, under a weighted distortion and a rank-deficient holder, has the convexity, pencil, and turning structure of the case without decoder side information is open. Explicit solutions are now available in three regimes: commuting geometry at every budget, arbitrary geometry in the one-dimensional regime at large budgets, and arbitrary geometry at small budgets when the decoder reproduces the whole source. For a partial description in non-commuting geometry between the two ends, the solution is a convex program over descriptions with at most rank Q components, and a closed form there, or for the regime boundaries, is not known. Stationary sources and finite blocklengths are not treated.

APPENDIX A THE CODING THEOREM

A. Finite alphabets

Let (X, S) be a discrete memoryless pair with finite alphabets and let d be a distortion with values in $[0, d_{\max}]$; codes, rate, distortion, and leakage $H(M | S^n)/n$ are as in Section II.

Theorem 26 ([1, Theorem 3], without decoder side information). *The closure of the achievable set at distortion D is $\bigcup_{p(\hat{x}|x): \mathbb{E}d(X, \hat{X}) \leq D} \{(R, L) : R \geq I(X; \hat{X}), L \geq I(X; \hat{X} | S)\}$, and the union is convex and closed.*

Proof outline. Villard and Piantanida prove the region in the equivocation form $\Delta \leq H(X | V) + 0 - I(X; S | U)$, $U - V - X - S$, with no decoder side information; taking $U = V$ is optimal, because $I(X; S | U) = I(V; S | U) + I(X; S | V) \geq I(X; S | V)$, and $\Delta = H(X | S) - L$ for a deterministic encoder, which gives the displayed form with V in place of $\hat{X}$, and a reproduction that is a function of V only lowers both coordinates. For the Gaussian extension we use the following finite-blocklength form of the achievability, obtained by covering at rate $I(X; \hat{X}) + 2\epsilon$, a random binning of the codeword index into $2^{n(I(X; \hat{X}|S) + 3\epsilon)}$ bins used only as a proof device, Fano's inequality for a hypothetical decoder holding the bin index and S^n , and selection of one deterministic code: for every channel with finite reproduction alphabet and every $\epsilon > 0$ there are $\epsilon_n \rightarrow 0$ and, for all large n , a deterministic code with $|\mathcal{M}_n| \leq 2^{n(I(X; \hat{X}) + 2\epsilon)}$, distortion at most $\mathbb{E}d(X, \hat{X}) + \epsilon + 2\epsilon_n d_{\max}$, and $H(M | S^n) \leq n(I(X; \hat{X} | S) + 3\epsilon) + 1 + 2\epsilon_n n(I(X; \hat{X}) + 2\epsilon)$. Convexity of both coordinates in the channel, for the conditional one by the decomposition $I(X; \hat{X} | S) = \sum_s p(s) I_{p(x|s)}(q)$, makes the union convex without time sharing. $\square$

B. The Gaussian source

Proof of Theorem 3. Converse. For any code, $\log |\mathcal{M}_n| \geq I(T^n; \hat{\mathbf{Y}}^n) \geq \sum_i I(T_i; \hat{\mathbf{Y}}_i)$ and $H(M | S^n) \geq I(T^n; \hat{\mathbf{Y}}^n | S^n) = \sum_i [I(T_i; \hat{\mathbf{Y}}^n, T^{i-1}, S^n) - I(T_i; S_i)] \geq \sum_i I(T_i; \hat{\mathbf{Y}}_i | S_i)$, by the chain rule and the independence of the pairs (T_i, S_i) ; every term is finite, being at most $I(T_i; S_i) + \log |\mathcal{M}_n|$ with $I(T_i; S_i) = \frac{1}{2} \log \det(H\Sigma_T H^\top + \Sigma_U) / \det \Sigma_U$. The mixture of the per-letter channels satisfies the Markov chain, has distortion D_n , and by convexity of both coordinates in the channel (for general alphabets the conditional coordinate is an integral over s of relative entropies affine in the channel [25], [24]) lies below (R_n, L_n) , so (R_n, L_n) is in the union at D_n .

Closedness. Let $(R_j, L_j) \rightarrow (R, L)$ lie in the union at levels D_j with $\limsup D_j \leq D$, with $\Sigma_j \in \mathcal{P}(D_j)$ below them by Proposition 2. The Σ_j lie in the compact set $0 \leq \Sigma \preceq \Sigma_T$ and $\det \Sigma_j \geq \det \Sigma_T 2^{-2R_j}$ is bounded away from zero, so a subsequence converges to some $\Sigma^* \succ 0$ in $\mathcal{P}(D)$; r and ℓ are continuous on positive definite matrices, so (R, L) lies above $(r(\Sigma^*), \ell(\Sigma^*))$.

Achievability. Fix an admissible channel with $I(T; \hat{\mathbf{Y}}) < \infty$; $\mathbb{E}\|\hat{\mathbf{Y}}\|^2 < \infty$ because its distortion is finite and $W \succ 0$. Let q_k be the saturating dyadic quantizer of level k applied coordinatewise: the bounded cells $[j2^{-k}, (j+1)2^{-k})$, $-4^k \leq j < 4^k$, map to their left endpoints and the two tail cells to $\pm 2^k$; the partitions are nested and generate the Borel σ -field.

Step A. The σ -fields of $S_\eta := q_\eta(S)$ increase to $\sigma(S)$, so $I(\hat{\mathbf{Y}}; S_\eta) \uparrow I(\hat{\mathbf{Y}}; S)$ [22, Ch. 2]; fix η with $I(\hat{\mathbf{Y}}; S_\eta) \geq I(\hat{\mathbf{Y}}; S) - \epsilon$. Since $\hat{\mathbf{Y}} - T - S_\eta$, $I(T; \hat{\mathbf{Y}} | S_\eta) = I(T; \hat{\mathbf{Y}}) - I(\hat{\mathbf{Y}}; S_\eta) \leq I(T; \hat{\mathbf{Y}} | S) + \epsilon$.

Step B. With $T_k = q_k(T)$ and $\hat{\mathbf{Y}}_k = q_k(\hat{\mathbf{Y}})$, the surrogate source (T_k, S_η) is discrete memoryless, the surrogate channel is $p(\hat{\mathbf{y}} | t_k) = \Pr(\hat{\mathbf{Y}}_k = \hat{\mathbf{y}} | T_k = t_k)$, and the surrogate distortion $d_k(t_k, \hat{\mathbf{y}}) = \mathbb{E}[(Y - \hat{\mathbf{y}})^\top W(Y - \hat{\mathbf{y}}) | T_k = t_k]$ is bounded for each k , every cell having positive probability since $\Sigma_T \succ 0$. The finite-blocklength achievability of Theorem 26 gives a code of rate at most $I(T_k; \hat{\mathbf{Y}}_k) + 2\epsilon \leq I(T; \hat{\mathbf{Y}}) + 2\epsilon$. Its true distortion equals its surrogate distortion, because given T_k^n the law of Y_i depends on $T_{k,i}$ alone. The surrogate distortion of the channel converges to $\mathbb{E}[(Y - \hat{\mathbf{Y}})^\top W(Y - \hat{\mathbf{Y}})] \leq D$, because $\mathbb{E}[Y | T_k] \rightarrow Y$ in L^2 by martingale convergence and $\hat{\mathbf{Y}}_k \rightarrow \hat{\mathbf{Y}}$ in L^2 . The surrogate law of $(\hat{\mathbf{Y}}_k, S_\eta)$ converges weakly to the true law of $(\hat{\mathbf{Y}}, S_\eta)$, by the bounded-convergence and martingale argument applied to test functions bounded on $\mathbb{R}^m \times q_\eta(\mathbb{R}^r)$ and continuous in the first argument, and the Markov chain $\hat{\mathbf{Y}} - T - S$. Mutual information is lower semicontinuous under weak convergence [23], so $I_k(T_k; \hat{\mathbf{Y}}_k | S_\eta) = I(T_k; \hat{\mathbf{Y}}_k) - I_k(\hat{\mathbf{Y}}_k; S_\eta) \leq I(T; \hat{\mathbf{Y}} | S) + 2\epsilon$ for large k , and the code has $H(M | S_\eta^n) \leq n(I(T; \hat{\mathbf{Y}} | S) + 5\epsilon + o(1))$.

Step C. S_η^n is a function of S^n , so $H(M | S^n) \leq H(M | S_\eta^n)$. A diagonal sequence $\epsilon \downarrow 0$ achieves the corner $(I(T; \hat{\mathbf{Y}}), I(T; \hat{\mathbf{Y}} | S))$ at distortion D , and with the converse, closedness, and Proposition 2 the region is as stated. $\square$

APPENDIX B

THE GAUSSIAN EXHAUSTION LEMMA

Proof of Lemma 1. The divergence bound. Conditional mutual information is the expected divergence of the conditional law of T given $(\hat{\mathbf{Y}}, S)$ from the conditional law given S [24], [22]. Fix a realization, let P be the

conditional law with mean μ and covariance C , $P_G = \mathcal{N}(\mu, C)$, and $Q_S = \mathcal{N}(\mu_S, \Sigma_{T|S})$. If P has no density the divergence is infinite. Otherwise $\log(dP_G/dQ_S)$ is quadratic in t and P, P_G share two moments, so $D(P\|Q_S) = D(P\|P_G) + D(P_G\|Q_S) \geq D(P_G\|Q_S)$. Averaging $D(P_G\|Q_S)$, the trace term plus the mean term, $\mathbb{E}[\text{tr}(\Sigma_{T|S}^{-1}C) + (\mu - \mu_S)^\top \Sigma_{T|S}^{-1}(\mu - \mu_S)]$, averages to p by the law of total covariance, and by concavity of $\ln \det$ and Jensen, $\mathbb{E}[-\ln \det C] \geq -\ln \det \mathbb{E}C \geq -\ln \det \Sigma_e$, since the conditional mean has error covariance no larger than any linear estimator. Hence $I(T; \hat{\mathbf{Y}} | S) \geq \frac{1}{2} \log(\det \Sigma_{T|S} / \det \Sigma_e)$, and the same argument without S gives $I(T; \hat{\mathbf{Y}}) \geq \frac{1}{2} \log(\det \Sigma_T / \det \Sigma_{e0})$.

The determinant identity. With $\hat{\mathbf{Y}} = AT + N'$, $\text{Cov}(N', T) = 0$, and $\text{Cov}(N', S) = 0$, the change of variables $(T, S, \hat{\mathbf{Y}}) \mapsto (T, S, N')$ is block triangular with unit diagonal, so $\det \text{Cov}(T, S, \hat{\mathbf{Y}}) = \det \Sigma_S \det \Sigma_{T|S} \det \Sigma_N$; the Schur factorization in the other order gives $\det \text{Cov}(\hat{\mathbf{Y}}, S) \det \Sigma_e$, and $\text{Cov}(\hat{\mathbf{Y}}) - \text{Cov}(\hat{\mathbf{Y}}, S) \Sigma_S^{-1} \text{Cov}(S, \hat{\mathbf{Y}}) = A \Sigma_{T|S} A^\top + \Sigma_N \succ 0$. So $\det \Sigma_e = \det \Sigma_{T|S} \det \Sigma_N / \det(A \Sigma_{T|S} A^\top + \Sigma_N)$, finite and positive.

Equality. For the jointly Gaussian channel $AT + N$ the conditional law of T given $(\hat{\mathbf{Y}}, S)$ is Gaussian with the linear estimate as mean and Σ_e as covariance, so every inequality is an equality; its second moments match the given channel's. $\square$

APPENDIX C NUMERICAL CHECKS

The identities and instances were checked independently of the derivations by scripts run on a remote workstation. `verify_general.py` checks, for random sources with general weights and observation maps, that the pencil value of Theorem 11 equals the brute-force optimum over Gaussian descriptions whenever that optimum is one-dimensional, that both costs are convex in the error covariance (16000 midpoint tests), that the weighted minimizers are unique, the two directions of Theorem 14, and the criterion of Theorem 16. `verify_commuting.py` checks the closed form of Theorem 8 for the minimum leakage and for weighted frontier points against brute force, the diagonality of the optimum in the common eigenbasis, the activation rule, and the block-diagonal optimum of Theorem 7. `verify_decoder_si.py` checks Lemma 23 for a non-Gaussian description (a sign quantizer, by numerical integration), the reduction of Theorem 24(b) on random Gaussian descriptions, and the minimum leakage with decoder side information against the reduced program. `verify_coupled.py`, `verify_parallel.py`, and `verify_pencil.py` check the scalar identity of Corollary 12 symbolically, the determinant program against brute force, the certificate, Lemma 20, the allocation, and the numbers of Section IX. `tqp_leakage_demo.py` computes Table I and Fig. 1. It evaluates the rate, distortion, and leakage of each quantizer by exact Gaussian integrals, checks that the leakage is strictly decreasing in $|\rho|$ for every quantizer, and compares the Lloyd–Max levels with the library tables of Section X. Its output is recorded in `tqp_leakage_demo-2026-10-01.txt`. `verify_fixed_read.py` checks the leakage of a fixed read against a direct conditional-covariance computation, the bounds of Proposition 18 and Theorem 19 on random instances against brute force over directions, and the numbers after Theorem 19; its output is recorded in `verify_fixed_read-2026-10-01.txt`. `verify_lowdist.py` checks Theorem 10 against the determinant program and a multi-start search on non-commuting instances, and the rank bound of Proposition 6; its output is recorded in `verify_lowdist-2026-10-01.txt`. No proof depends on them; the instances illustrate theorems proved in the text.

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